

The Sunlight Factory

Bonus · Photosynthesis in Higher Plants

Starring Chloro the leaf factory 🏭

A BAIO storybook by Narendra Sir · Biology Lecturer



The Factory Is Failing

In a green leaf, a tiny factory named Chloro was running out of power. "My machines need three things: **sunlight**, **water** and **carbon dioxide**. Without them the whole plant will starve!" Chloro's job is **photosynthesis**: $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$, using light energy. Students, you must help collect the ingredients!

💡 **REMEMBER**

Photosynthesis is the basis of life on Earth: it gives us food and releases oxygen into the atmosphere.



Gates and Pipes

First, carbon dioxide entered through tiny pores on the leaf called **stomata**. Water climbed all the way from the roots through the **xylem**. Long ago **Joseph Priestley** showed that a plant could restore air ‘spoiled’ by a burning candle, and **Jan Ingenhousz** showed that this happened only in sunlight. Later it was proved that the oxygen released comes from **water**, not from CO_2.

💡 **REMEMBER**

Cornelius van Niel showed that hydrogen from water reduces CO_2 to carbohydrate; oxygen comes from water.



The Solar Panels

Inside the **chloroplast**, stacks of **thylakoids** called grana held the solar panels — pigments. **Chlorophyll a**, the chief pigment, absorbs mainly blue and red light. **Chlorophyll b**, **xanthophylls** and **carotenoids** are **accessory pigments** that widen the range of light used and protect chlorophyll a. Pigments are grouped in **light harvesting complexes** around a reaction centre.

💡 **REMEMBER**

PS I has the reaction centre P700 and PS II has P680, named after the wavelengths they absorb best.



The Light Reaction

When light struck **PS II**, electrons were boosted to a high energy level. To replace them, water was split — **photolysis** — releasing oxygen bubbles. Electrons flowed along the **Z scheme** to **PS I** and finally reduced $NADP^+$ to **NADPH**. Protons built up inside the thylakoid and flowed out through ATP synthase to make **ATP** — **chemiosmosis**.

💡 **REMEMBER**

Making ATP using light is photophosphorylation; cyclic photophosphorylation uses only PS I and makes no NADPH or O_2.



The Sugar Kitchen

ATP and NADPH carried energy to the **stroma**, where the **Calvin cycle** ran. The enzyme **RuBisCO** joined CO_2 to **RuBP** — **carboxylation** — making two molecules of **3-PGA**, the first stable product. So these plants are called **C_3 plants**. Next came reduction to sugar and regeneration of RuBP.

💡 **REMEMBER**

To make one glucose molecule the Calvin cycle turns 6 times, using 18 ATP and 12 NADPH.



The C_4 Shortcut

In a hot maize field, Chloro met a cousin. "We are **C_4 plants** like maize and sugarcane. Our first product is the four-carbon **oxaloacetic acid**." In the mesophyll, **PEP carboxylase** fixes CO_2; then the Calvin cycle runs inside large **bundle sheath cells** arranged in a wreath — **Kranz anatomy**. "We do not suffer from photorespiration!"

💡 **REMEMBER**

C_4 plants tolerate high temperatures, show higher productivity and respond to high light intensities.



A Wasteful Mistake

On hot, bright days RuBisCO sometimes grabbed O_2 instead of CO_2 — **photorespiration**. No sugar or ATP was made; energy was wasted. Chloro also learnt **Blackman's law of limiting factors**: when many factors affect a process, the rate is set by the one in shortest supply — light, **CO_2 concentration**, **temperature** or water.

💡 **REMEMBER**

CO_2 is the major limiting factor; tomato and bell pepper greenhouses are enriched with CO_2 for higher yields.



Glucose and Oxygen

With sunlight, water and CO_2 flowing in, Chloro's machines hummed. **Glucose** was made for food and stored as starch, and **oxygen** bubbled out of the stomata for every animal to breathe. The plant grew strong and green again. "Every bite you eat," said Chloro proudly, "began as sunlight caught in a leaf."

💡 **REMEMBER**

Products of photosynthesis: glucose (carbohydrate) and oxygen; water is also formed.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

sunlight p.1 · water p.1 · carbon dioxide p.1 · photosynthesis p.1

stomata p.2 · xylem p.2 · Joseph Priestley p.2 · Jan Ingenhousz p.2

chloroplast p.3 · thylakoids p.3 · Chlorophyll a p.3 · Chlorophyll b p.3

xanthophylls p.3 · carotenoids p.3 · accessory pigments p.3

light harvesting complexes p.3 · PS II p.4 · photolysis p.4

Z scheme p.4 · PS I p.4 · NADPH p.4 · ATP p.4 · chemiosmosis p.4

stroma p.5 · Calvin cycle p.5 · RuBisCO p.5 · RuBP p.5

carboxylation p.5 · 3-PGA p.5 · C_3 plants p.5 · C_4 plants p.6

oxaloacetic acid p.6 · PEP carboxylase p.6 · bundle sheath cells p.6

Kranz anatomy p.6 · photorespiration p.7

Blackman's law of limiting factors p.7 · CO_2 concentration p.7

temperature p.7 · Glucose p.8 · oxygen p.8



QUICK QUIZ

Do you remember?

1. What gas do plants take in for photosynthesis?

- Oxygen
- Carbon dioxide
- Nitrogen
- Hydrogen

2. Which pigment is chiefly responsible for absorbing sunlight?

- Chlorophyll a
- Xanthophyll
- Haemoglobin
- Carotene

3. What are the products of photosynthesis?

- CO_2 and water
- Glucose and oxygen
- ATP only
- Starch and CO_2

4. The first stable product of the Calvin cycle is:

- Oxaloacetic acid
- 3-PGA
- RuBP
- Glucose

5. The oxygen released in photosynthesis comes from:

- CO_2
- Water
- Glucose
- Air

Answers: 1-Carbon dioxide · 2-Chlorophyll a · 3-Glucose and oxygen · 4-3-PGA · 5-Water